

Multiple Choice (10 marks)

1. In Yann LeCun's cake, the cherry on top is
 - (a) reinforcement learning
 - (b) self-supervised learning
 - (c) unsupervised learning
 - (d) supervised learning
2. Which one of the following is the main reason for pruning a Decision Tree?
 - (a) To save computing time during testing
 - (b) To save space for storing the Decision Tree
 - (c) To make the training set error smaller
 - (d) To avoid overfitting the training set
3. High entropy means that the partitions in classification are
 - (a) pure
 - (b) not pure
 - (c) useful
 - (d) useless
4. Statement 1| The softmax function is commonly used in mutliclass logistic regression.
Statement 2| The temperature of a nonuniform softmax distribution affects its entropy.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
5. Which one of the following is equal to $P(A, B, C)$ given Boolean random variables A , B and C , and no independence or conditional independence assumptions between any of them?
 - (a) $P(A | B) * P(B | C) * P(C | A)$
 - (b) $P(C | A, B) * P(A) * P(B)$
 - (c) $P(A, B | C) * P(C)$
 - (d) $P(A | B, C) * P(B | A, C) * P(C | A, B)$

True / False (10 marks)

Answer True or False

6. True or False: L 2 regularization of linear models tends to produce sparser models than L 1 regularization, enhancing feature selection.
7. True or False: The softmax function is commonly employed in multiclass logistic regression to model probability distributions over multiple classes.
8. True or False: Averaging the output of multiple decision trees increases bias in model predictions.
9. True or False: Grid search runs reasonably slow for multiple linear regression due to the extensive parameter combinations it evaluates.
10. True or False: The method used to learn weights, such as matrix inversion or gradient descent, primarily affects the trade-off between underfitting and overfitting.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the process of calculating the gradient of the objective function $\frac{1}{2}Xw - y_2^2 + \frac{1}{2}\lambda w_2^2$ with respect to w , and explain its significance in optimization.
12. Given two Boolean random variables A and B with specified probabilities, derive $P(A | B)$ and discuss the implications of this relationship in the context of conditional probability in machine learning.

End of Paper